Mathematics practice booklet: Grade 3 expectations

Grade 3 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 3 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 3.OA.A.1, 3.OA.A.2, 3.OA.A.3, 3.OA.A.4, 3.OA.B.5, 3.OA.B.6, 3.OA.C.7, 3.OA.D.8, 3.OA.D.9

A worked example

There are 6 trays with 4 date cookies on each tray. How many cookies are there? Write the multiplication fact and a related division fact.

1. 6 groups of 4 is 6 × 4.
2. 6 × 4 = 24, so there are 24 cookies.
3. A related division fact: 24 / 6 = 4.

24 cookies; 6 × 4 = 24 and 24 / 6 = 4

Questions

1. What does 5 × 7 mean? Say it as groups.

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2. A box holds 8 pencils. How many pencils are in 4 boxes?

   A. 12

   B. 24

   C. 32

   D. 36

3. 30 stickers are shared equally among 5 friends. How many does each friend get?

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4. Hamad packs 36 date cookies, 4 on each tray. How many trays does he fill?

   A. 8

   B. 9

   C. 10

   D. 32

5. Find the missing number: 42 / __ = 6

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6. Use a fact you know to find 6 × 8. Hint: 6 × 8 is the same as 8 × 6, and 5 × 8 = 40.

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7. Are 3 × 9 and 9 × 3 equal? Which property shows it?

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8. Maha has 3 packs of 6 stickers and gives 5 away. How many does she have left?

   A. 13

   B. 18

   C. 23

   D. 8

9. What is the rule for this pattern, and what comes next? 4, 8, 12, 16, __

   A. Add 2; the next number is 18

   B. Multiply by 2; the next number is 32

   C. Add 4; the next number is 19

   D. Add 4; the next number is 20

10. Write a number sentence for: a number times 7 is 56. Use n for the unknown, then solve.

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11. A baker packs 24 date cookies into boxes of 6. How many boxes does she fill? Write the division sentence.

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12. Fluency check: 7 × 8 = ? and 63 / 9 = ?

Section 2: Shapes, measuring and space

Standards practised: 3.MD.C.5, 3.MD.C.6, 3.MD.C.7, 3.MD.D.8, 3.G.A.1, 3.G.A.2, 3.MD.A.1, 3.MD.A.2

A worked example

A rectangular majlis rug is 4 metres long and 3 metres wide. What is its area? What is its perimeter?

1. Area is the number of square metres that cover it: 4 × 3 = 12.
2. Perimeter is the distance round the edge: 4 + 3 + 4 + 3 = 14.

Area 12 square metres; perimeter 14 metres

Questions

1. A rug in a model of a majlis is 5 cm long and 2 cm wide. What is the area of the rug?

   A. 7 square centimetres

   B. 10 square centimetres

   C. 14 square centimetres

   D. 10 centimetres

2. Layla makes a square card with sides of 6 cm and sticks a ribbon all the way round the edge. How long is the ribbon?

   A. 12 cm

   B. 24 cm

   C. 36 cm

   D. 6 cm

3. A rectangle has area 24 square units and a width of 4 units. What is its length?

   _____

4. Name a shape with 4 sides and 4 equal sides. Is it always a square?

   _____

5. A shape is split into 4 equal parts. What fraction of the whole is each part?

   _____

6. The film starts at 4:35 and lasts 40 minutes. When does it end?

   A. 5:05

   B. 5:25

   C. 5:15

   D. 5:35

7. A rectangle has sides 7 cm and 3 cm. Find its perimeter.

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8. Draw two different rectangles that both have an area of 12 square units. What are their sides?

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9. A jug holds 2 litres of juice. Hamad pours out 500 millilitres. How many millilitres are left? (1 litre = 1000 ml)

   A. 500 ml

   B. 1,000 ml

   C. 2,500 ml

   D. 1,500 ml

10. Layla has two cards. One is a 5 by 3 rectangle and the other is a 4 by 4 square. Which has the larger area?

    A. The 5 by 3 rectangle

    B. The 4 by 4 square

    C. They are equal

    D. You cannot tell

Section 3: Data, graphs and chance

Standards practised: 3.MD.B.3, 3.MD.B.4

A worked example

A bar graph shows books read in a month. Each square stands for 2 books. Salem has 5 squares and Mona has 3 squares. How many more books did Salem read?

1. Salem read 5 × 2 = 10 books.
2. Mona read 3 × 2 = 6 books.
3. 10 - 6 = 4.

4 more books

Questions

1. In a picture graph each star is 5 votes. A drink has 4 stars. How many votes?

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2. In a picture graph each star is 5 votes. Juice has 6 stars and milk has 4 stars. How many more votes does juice have?

   A. 2

   B. 20

   C. 10

   D. 30

3. A bar graph's scale goes up in 10s. A bar ends halfway between 20 and 30. What value does it show?

   A. 25

   B. 15

   C. 30

   D. 35

4. Pencil lengths in cm are 12, 14, 12, 15, 14, 12. How many pencils are 12 cm?

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5. Use the lengths in the last question. What is the longest length, and what is the shortest?

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6. A line plot shows lengths to the nearest half centimetre: 2 dots at 3 cm and 4 dots at 3 and a half cm. How many measurements are there?

   A. 2

   B. 7

   C. 4

   D. 6

7. A survey: 8 children like football, 5 like swimming, 7 like running. How many children in all?

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8. Which sport in that survey is the most popular, and by how many votes is it ahead of the least popular?

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9. A graph uses 1 square for every 3 students. 12 students walk to school. How many squares does the graph need?

   A. 3

   B. 4

   C. 6

   D. 9

10. Make a tally of the vowels a, e, i, o, u in the word CALENDAR. Which vowel shows the most, and how many times?

Section 4: Numbers and calculating

Standards practised: 3.NBT.A.1, 3.NBT.A.2, 3.NBT.A.3, 3.NF.A.1, 3.NF.A.2, 3.NF.A.3

A worked example

Round 487 to the nearest ten and to the nearest hundred.

1. Look at the ones digit for the nearest ten: 7 is 5 or more, so round up: 490.
2. Look at the tens digit for the nearest hundred: 8 is 5 or more, so round up: 500.

490 and 500

Questions

1. A shop rounds every price to the nearest ten dirhams. What does it make AED 364?

   A. AED 360

   B. AED 370

   C. AED 300

   D. AED 400

2. Round 751 to the nearest hundred.

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3. Salem has 346 date stones and finds 278 more. How many does he have now?

   A. 524

   B. 614

   C. 624

   D. 634

4. A shop has 503 stickers and sells 217. How many stickers are left?

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5. Maryam buys 6 packs of dates with 40 dates in each pack. How many dates is that in all?

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6. A bag holds 125 dates and a second bag holds 238. How many in total?

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7. What fraction of the circle is shaded if 3 of 8 equal parts are shaded?

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8. Maha eats 1/2 of a pizza and Noura eats 1/4 of the same pizza. Who eats more?

   A. Maha

   B. Noura

   C. They eat the same

   D. You cannot tell

9. Which fraction is equal to 1/2 and has 4 as its denominator?

   A. 1/4

   B. 3/4

   C. 4/4

   D. 2/4

10. Place 3/4 on a number line from 0 to 1. At which quarter mark do you stop?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 5 groups of 7. Multiplication means equal groups: 5 × 7 is 7 + 7 + 7 + 7 + 7 = 35.
2. C. 32. 4 boxes of 8 pencils is 4 × 8 = 32. If your child chose A (12), they added 4 + 8 instead of multiplying; if they chose D (36), they mixed it up with 4 × 9.
3. 6. Sharing equally means dividing: 30 / 5 = 6, because 5 × 6 = 30.
4. B. 9. Ask what times 4 makes 36: 4 × 9 = 36, so 36 / 4 = 9. If your child chose D (32), they subtracted 36 - 4 instead of dividing.
5. 7. Ask what times 6 makes 42: 6 × 7 = 42, so 42 / 7 = 6.
6. 48. Split it into facts you know: 6 × 8 = 5 × 8 + 1 × 8 = 40 + 8 = 48.
7. Yes, both are 27 (3 × 9 = 27 and 9 × 3 = 27). The order does not change the product, which is the commutative property.
8. A. 13. First find the stickers: 3 × 6 = 18. Then take away 5: 18 - 5 = 13. If your child chose C (23), they added 5; if they chose B (18), they forgot to take away the 5.
9. D. Add 4; the next number is 20. Each number is 4 more than the last, so 16 + 4 = 20. If your child chose B, they saw 4 to 8 doubles, but 8 to 12 does not.
10. n = 8. Write n x 7 = 56, then divide: 56 / 7 = 8, because 8 × 7 = 56.
11. 4 boxes. Count how many 6s fit into 24: 24 / 6 = 4, because 4 × 6 = 24.
12. 56 and 7. 7 × 8 = 56, and 63 / 9 = 7 because 9 × 7 = 63.

Section 2: Shapes, measuring and space

1. B. 10 square centimetres. Area is length x width: 5 × 2 = 10. If your child chose C (14), they added all four sides; if they chose A (7), they added 5 + 2.
2. B. 24 cm. The ribbon goes round all 4 sides: 4 × 6 = 24. If your child chose D (6 cm), they counted one side; if they chose C (36 cm), they found the area 6 × 6.
3. 6 units. Area is length x width, so 24 = length x 4, and 24 / 4 = 6.
4. A rhombus or a square, but not always a square. A rhombus can lean over, so its corners need not be right angles.
5. 1/4. Four equal parts are fourths, so each part is 1 out of 4, written 1/4.
6. C. 5:15. 4:35 plus 25 minutes is 5:00, and 15 more minutes is 5:15. If your child chose A (5:05), they added 30 minutes; if they chose D (5:35), they added a whole hour.
7. 20 cm. Opposite sides match, so add all four: 7 + 3 + 7 + 3 = 20.
8. 3 by 4 and 2 by 6, because 3 × 4 = 12 and 2 × 6 = 12. 1 by 12 works too.
9. D. 1,500 ml. Change to millilitres first: 2 litres = 2,000 ml, and 2,000 - 500 = 1,500. If your child chose C (2,500 ml), they added 500; if they chose A (500 ml), they gave the amount poured out.
10. B. The 4 by 4 square. 5 × 3 = 15 and 4 × 4 = 16, so the square is larger. If your child chose A, they compared the long sides, 5 and 4, instead of the areas.

Section 3: Data, graphs and chance

1. 20. Each star is 5 votes, so 4 stars is 4 × 5 = 20 votes.
2. C. 10. Juice has 6 - 4 = 2 more stars, and 2 × 5 = 10 votes. If your child chose A (2), they forgot that each star is worth 5.

3. A. 25. Halfway between 20 and 30 is 25, because 20 + 5 = 25. If your child chose C (30), they picked the next mark instead of the middle.
4. 3. Go through the list and count each 12: the 1st, 3rd and 6th numbers, so 3 pencils.
5. Longest 15 cm, shortest 12 cm. Find the biggest and the smallest numbers in the list.
6. D. 6. Each dot is one measurement, so add the dots: 2 + 4 = 6. If your child chose C (4), they counted only the tallest stack of dots.
7. 20. Add the three groups: 8 + 5 = 13, then 13 + 7 = 20.
8. Football, with 8 votes. It is ahead of swimming, with 5 votes, by 8 - 5 = 3.
9. B. 4. Each square stands for 3 students, so divide: 12 / 3 = 4. If your child chose D (9), they subtracted 12 - 3 instead of dividing.
10. A shows the most. CALENDAR has A twice and E once, so A appears 2 times and E 1 time.

Section 4: Numbers and calculating

1. A. AED 360. The ones digit 4 is less than 5, so round down. If your child chose B (AED 370), they rounded up; if they chose C (AED 300), they rounded to the nearest hundred.
2. 800. Look at the tens digit: 5 is 5 or more, so 751 rounds up to 800.
3. C. 624. Add by place: 500 + 110 + 14 = 624. If your child chose B (614), they lost the ten from 6 + 8 = 14.
4. 286. Subtract in steps: 503 - 200 = 303, then 303 - 17 = 286.
5. 240. 6 × 4 tens = 24 tens, and 24 tens is 240.
6. 363. Add by place: 100 + 200 = 300, 20 + 30 = 50, 5 + 8 = 13, and 300 + 50 + 13 = 363.
7. 3/8. The circle has 8 equal parts and 3 are shaded, so the fraction is 3 out of 8.
8. A. Maha. A half is bigger than a quarter, because 2 quarters make a half. If your child chose B (Noura), they thought the bigger bottom number meant a bigger piece.
9. D. 2/4. Two of four equal parts make a half, so 2/4 = 1/2. If your child chose A (1/4), they kept the top number 1 and only changed the bottom.
10. At the third mark. Split 0 to 1 into 4 equal parts; 3/4 is 3 parts from 0, one mark before 1.